

Project: Coffee Shop Sales Analysis

Problem Statement

A coffee shop wants to analyze the sales performance of its different coffee drinks. The owner has collected data about the number of cups sold and the price of each coffee. As a Data Analyst, your task is to use NumPy and Pandas to organize the data, perform basic analysis, and generate a simple sales report.

Objective

Create a Python program using NumPy and Pandas to analyze the coffee shop sales data and display meaningful business insights.

Requirements

1. Import the NumPy and Pandas libraries.
2. Create NumPy arrays for the following information:
 - Coffee Name
 - Cups Sold
 - Price per Cup
3. Create a Pandas DataFrame using the NumPy arrays.
4. Display the complete DataFrame.
5. Create a new column named 'Revenue'.
 $\text{Revenue} = \text{Cups Sold} \times \text{Price per Cup}$
6. Display the updated DataFrame with the Revenue column.
7. Find and display the following:
 - Most Sold Coffee
 - Most Expensive Coffee
 - Average Coffee Price
 - Total Revenue Earned
8. Display the final sales report in a neat and readable format.

Sample Data

Coffee	Cups Sold	Price (₹)
Espresso	120	120
Latte	95	180
Cappuccino	150	160
Mocha	80	200
Cold Coffee	110	150

Concepts Covered

- Importing Libraries
- NumPy Arrays
- Pandas DataFrame
- Creating New Columns
- Basic Statistical Functions
- Finding Maximum Values
- Calculating Sum and Mean
- Displaying Business Reports

Expected Learning Outcomes

After completing this project, students will be able to:

- Create NumPy arrays.
- Build a Pandas DataFrame.
- Add new calculated columns.
- Perform simple business analysis.
- Generate a basic analytical report using Python.